Sphinx Tutorial

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This tutorial demonstrates a number of Sphinx features useful for technical documentation.

CHAPTER

ONE

CODE LISTING

1.1 Syntax highlighting

Some of the useful code-listing features are:

- syntax highlighting
- line numbering
- line highlighting (emphasis)
- including external files.

Highlighting of programming languages are supported by Pygments.

Pygments has a collection of syntax highlighting styles:

```
from pygments.styles import get_all_styles
print(list(get_all_styles()))
```

['default', 'emacs', 'friendly', 'colorful', 'autumn', 'murphy', 'manni', 'monokai', 'perldoc', 'pastie', 'borland', 'trac', 'native', 'fruity', 'bw', 'vim', 'vs', 'tango', 'rrt', 'xcode', 'igor', 'paraiso-light', 'paraiso-dark', 'lovelace', 'algol', 'algol_nu', 'arduino', 'rainbow_dash', 'abap']

For example, in `conf.py`:

```
pygments_style = 'emacs'
```

1.2 Examples

1.2.1 Inlined listing

The code listing is embedded in the text:

Bash:

```
export PATH=$HOME/bin:$PATH
for i in `seq 1 10`; do
   echo $i
   echo $LD_LIBRARY_PATH
done
```

C language (also with line emphasis):

```c
#include <stdio.h>

int static hello(int *i)
{
   double *x;
   return 0;
}

for (int i=0; i<10; i++) {
  printf("i = %d\n", i);
}
```

1.2.2 Including external file(s)

The code is in an external file, e.g. `foo.py`:

```python
import matplotlib
import numpy

def hello():
    print 'hello world'

if __name__ == '__main__':
    hello()
```

Code listing without highlighting:

```
import matplotlib
import numpy

def hello():
    print 'hello world'

if __name__ == '__main__':
    hello()
```

CHAPTER

TWO

MATH

2.1 Equations written in LaTeX

Some equation examples:

$$\frac{\partial f(x,y)}{\partial x} = 0$$

$$\mathcal{F}(x) = \sum_{i=0}^{N} a_i \exp(ikx)$$

Inline math is also possible, e.g. $\sum_{i=0}^{N} a_i = 0$.

CHAPTER THREE

IMAGES

Notice that the file paths to the images are relative to the directory of this file.

3.1 Embedded Image

3.2 Figure

Fig. 1: This is the caption of the figure. The caption can be very long.

CHAPTER

FOUR

LINKS

4.1 Custom Anchor

Referencing an anchor point: *External Web page*

The reference text automatically changes if the target text is changed. (This is a good thing.)

4.2 External Web page

An external web page example: google search

Or google_search

CHAPTER

FIVE

MARKDOWN SUPPORT

5.1 Using Markdown

5.1.1 Markdown in Sphinx

Markdown files can be used in Sphinx. This page is written in markdown.

5.2 Code listing example

C code listing:

```
#include <stdio.h>
int main(int argc, char *argv[])
{
    printf("hello world\n");
    return 0;
}
```

5.3 Bullet points

- Markdown may be easier to write than reStructuredText
- But some Sphinx features are not supported
  - minor point 1
  - minor point 2

5.4 Reference

https://www.sphinx-doc.org/en/master/usage/markdown.html

CHAPTER

SIX

JUPYTER NOTEBOOK SUPPORT

Jupyter notebook files (`*.ipynb`) can be directly included in Sphinx by using the nbsphinx extension.

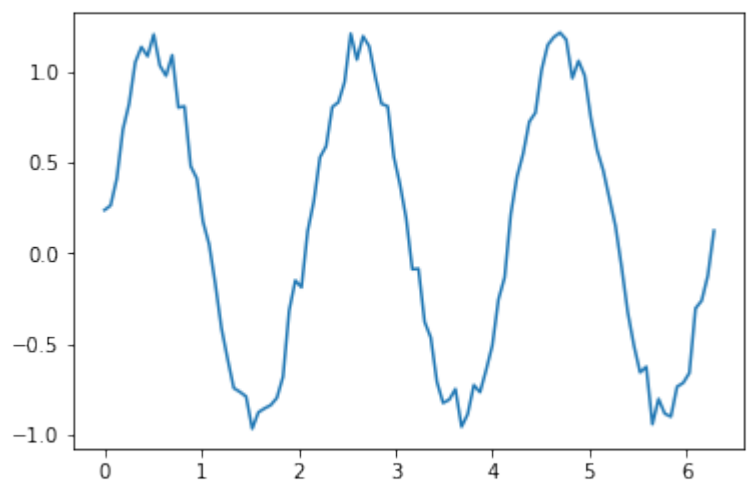
6.1 Jupyter notebook

6.1.1 A simple plot

[1]:
```
%matplotlib inline
import numpy as np
import matplotlib.pyplot as plt

N = 100
x = np.linspace(0,2*np.pi,N)
f = np.sin(3*x) + 0.25*np.random.random(N)
plt.plot(x,f,'-')
```

[1]:
```
[<matplotlib.lines.Line2D at 0x7f323aee6e10>]
```



6.1.2 Retrieving the web page

[2]:
```
import urllib.request
response = urllib.request.urlopen('http://www.ucla.edu/')
html = response.read()
```

CHAPTER

SEVEN

SPHINX CHEAT SHEET

(Originally from https://matplotlib.org/sampledoc/cheatsheet.html)

Here is a quick and dirty cheat sheet for some common stuff you want to do in sphinx and ReST. You can see the literal source for this file at *This file*.

7.1 Formatting text

You use inline markup to make text *italics*, **bold**, or `monotype`.

You can represent code blocks fairly easily:

```
import numpy as np
x = np.random.rand(12)
```

Or literally include code:

```
from pylab import *
from matplotlib.patches import Ellipse

delta = 45.0 # degrees

angles = arange(0, 360+delta, delta)
ells = [Ellipse((1, 1), 4, 2, a) for a in angles]

a = subplot(111, aspect='equal')

for e in ells:
    e.set_clip_box(a.bbox)
    e.set_alpha(0.1)
    a.add_artist(e)

xlim(-2, 4)
ylim(-1, 3)

show()
```

7.2 Making a list

It is easy to make lists in rest

7.2.1 Bullet points

This is a subsection making bullet points

- point A
- point B
- point C

7.2.2 Enumerated points

This is a subsection making numbered points

1. point A
2. point B
3. point C

7.3 Making a table

This shows you how to make a table – if you only want to make a list see *Making a list*.

Name	Age
John D Hunter	40
Cast of Thousands	41
And Still More	42

7.4 Making links

It is easy to make a link to yahoo or to some section inside this document (see *Making a table*) or another document.

You can also reference classes, modules, functions, etc that are documented using the sphinx autodoc facilites. For example, see the module `matplotlib.backend_bases` documentation, or the class `LocationEvent`, or the method `mpl_connect()`.

7.5 This file

```
.. _cheat-sheet:


******************
Sphinx cheat sheet
******************

(Originally from https://matplotlib.org/sampledoc/cheatsheet.html)

Here is a quick and dirty cheat sheet for some common stuff you want
to do in sphinx and ReST.  You can see the literal source for this
file at :ref:`cheatsheet-literal`.


.. _formatting-text:

Formatting text
===============

You use inline markup to make text *italics*, **bold**, or ``monotype``.

You can represent code blocks fairly easily::

   import numpy as np
   x = np.random.rand(12)

Or literally include code:

.. literalinclude:: src/ellipses.py

.. _making-a-list:

Making a list
=============

It is easy to make lists in rest

Bullet points
-------------

This is a subsection making bullet points

* point A

* point B

* point C


Enumerated points
------------------

This is a subsection making numbered points

#. point A
```

(continues on next page)

(continued from previous page)

```
#. point B

#. point C


.. _making-a-table:

Making a table
==============

This shows you how to make a table -- if you only want to make a list see :ref:
↪`making-a-list`.

==================   ============
Name                 Age
==================   ============
John D Hunter        40
Cast of Thousands    41
And Still More       42
==================   ============

.. _making-links:

Making links
============

It is easy to make a link to `yahoo <http://yahoo.com>`_ or to some
section inside this document (see :ref:`making-a-table`) or another
document.

You can also reference classes, modules, functions, etc that are
documented using the sphinx `autodoc
<http://sphinx.pocoo.org/ext/autodoc.html>`_ facilites.  For example,
see the module :mod:`matplotlib.backend_bases` documentation, or the
class :class:`~matplotlib.backend_bases.LocationEvent`, or the method
:meth:`~matplotlib.backend_bases.FigureCanvasBase.mpl_connect`.



.. _cheatsheet-literal:

This file
=========

.. literalinclude:: cheatsheet.rst
```

The PDF version of this site is also avaiable in a report format.